RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Friday, 11 September 2026, 4:03 PM
State	Finished
Completed on	Friday, 11 September 2026, 4:04 PM
Time taken	1 min 4 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findDuplicate(const int* nums, int n) {
    int slow = nums[0];
    int fast = nums[0];

    // Phase 1: Finding the intersection point in the cycle
    do {
        slow = nums[slow];
        fast = nums[nums[fast]];
    } while (slow != fast);

    // Phase 2: Finding the entrance to the cycle (the duplicate number)
    slow = nums[0];
    while (slow != fast) {
        slow = nums[slow];
        fast = nums[fast];
    }

    return slow;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *nums = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    int duplicate = findDuplicate(nums, n);
    printf("%d\n", duplicate);

    free(nums);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔

	Input	Expected	Got	
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Friday, 11 September 2026, 3:31 PM
State	Finished
Completed on	Friday, 11 September 2026, 3:33 PM
Time taken	1 min 54 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int slow = arr[0];
    int fast = arr[0];

    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);

    fast = arr[0];
    while (slow != fast) {
        slow = arr[slow];
        fast = arr[fast];
    }

    printf("%d\n", slow);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Friday, 11 September 2026, 3:33 PM
State	Finished
Completed on	Friday, 11 September 2026, 3:33 PM
Time taken	38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

void solve() {
    int n1, n2;
    if (scanf("%d", &n1) != 1) return;
    int *arr1 = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    if (scanf("%d", &n2) != 1) {
        free(arr1);
        return;
    }
    int *arr2 = (int *)malloc(n2 * sizeof(int));
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1;
    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
        } else if (arr2[j] < arr1[i]) {
            j++;
        } else {
            if (!first) {
```

```
            printf(" ");
        }
        printf("%d", arr1[i]);
        first = 0;
        i++;
        j++;
    }
  }
  printf("\n");

  free(arr1);
  free(arr2);
}

int main() {
  int t;
  if (scanf("%d", &t) == 1) {
    while (t--) {
      solve();
    }
  }
  return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Friday, 11 September 2026, 3:33 PM
State	Finished
Completed on	Friday, 11 September 2026, 3:34 PM
Time taken	30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

void solve() {
    int n1, n2;
    if (scanf("%d", &n1) != 1) return;
    int *arr1 = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    if (scanf("%d", &n2) != 1) {
        free(arr1);
        return;
    }
    int *arr2 = (int *)malloc(n2 * sizeof(int));
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1;
    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
        } else if (arr2[j] < arr1[i]) {
            j++;
        } else {
            if (!first) {
```

```
            printf(" ");
        }
        printf("%d", arr1[i]);
        first = 0;
        i++;
        j++;
    }
  }
  printf("\n");

  free(arr1);
  free(arr2);
}

int main() {
  int t;
  if (scanf("%d", &t) == 1) {
    while (t--) {
      solve();
    }
  }
  return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Lathika V H 251001132-2025-IT

L2

Started on	Friday, 11 September 2026, 3:34 PM
State	Finished
Completed on	Friday, 11 September 2026, 3:35 PM
Time taken	23 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0, j = 1;
    int found = 0;

    while (i < n && j < n) {
        if (i != j && arr[j] - arr[i] == k) {
            found = 1;
            break;
        } else if (arr[j] - arr[i] < k) {
            j++;
        } else {
            i++;
        }
    }

    printf("%d\n", found);

    free(arr);
    return 0;
}

```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

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L2

Started on	Friday, 11 September 2026, 3:35 PM
State	Finished
Completed on	Friday, 11 September 2026, 3:35 PM
Time taken	24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0, j = 1;
    int found = 0;

    while (i < n && j < n) {
        if (i != j && arr[j] - arr[i] == k) {
            found = 1;
            break;
        } else if (arr[j] - arr[i] < k) {
            j++;
        } else {
            i++;
        }
    }

    printf("%d\n", found);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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